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## **An Adaptive Droop Method for Local Reactive Power Compensation in an MV Microgrid**

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### **SUMMARY**

A natural evolution of the smart grid concept is the transition from the conventional distribution system to an active distribution system (ADS). A microgrid is a cluster of distributed generation (DG) units and a distribution network that provides the reliable operation for the critical loads, mainly when the grid supply is not present. This paper presents an adaptive droop scheme for local reactive power compensation of a medium voltage (MV) microgrid. The microgrid consists of several electronically-interfaced Distributed Generation (DG) units. The proposed technique provides reactive power sharing even for the nonlocal loads among the DG units. Therefore, it can be integrated with the existing voltage control methods. The principles of operation and the effectiveness of the proposed method are discussed. Moreover, the performance of the method is verified based on time-domain simulation studies carried out in the PSCAD/EMTDC software.

### **KEYWORDS**

Adaptive droop, distributed generation, microgrid, local reactive power compensation.

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## 1. INTRODUCTION

Microgrids and smart grids are the new concepts that have been brought about due to high penetration of renewable energy systems, DG units, and storage systems [1-6]. The environmental issues and economical, social, and political interests make the role of microgrid even more important. The recently proposed concept of multi-microgrids corresponds to a higher level structure formed at a Medium Voltage (MV) level. A multi-microgrid consists of Low Voltage (LV) microgrids and DG units which are connected to several adjacent MV feeders [2]. Microgrids, DG units and MV loads under the demand-side management control can be considered as active cells for control and management purposes. In an MV microgrid, a coordinating procedure to control reactive power flow and to ultimately control the grid voltage is required [7]. The procedure should minimize the losses and the other relevant costs. Moreover, to avoid the serious problems, e.g., high losses of the overall microgrid and system instability, the coordinating procedure should play the role of reactive power management [3].

This paper proposes an adaptive decentralized control scheme for local compensation of reactive power in an MV microgrid. The proposed method does not require any communication link and minimizes the reactive power flow in the MV lines; thus reducing the losses of the overall microgrid. The ability of the method in compensating the local reactive power in an islanded microgrid is verified by the simulation studies carried out in PSCAD/EMTDC.

## 2. MULTI-BUS MV MICROGRID STRUCTURE AND DESCRIPTION OF CONTROL SYSTEM

Fig. 1 shows a single-line diagram of a multi-bus MV microgrid consisting of a 20-kV three-feeder distribution system and two electronically-coupled three-wire DG units. The DG1 and DG2 are respectively connected to the end of feeders F1 and F2 through step up transformers and are assumed dispatchable. Thus, each DG unit can supply any amount of the real/reactive power within the pre-specified limits. Moreover, each DG unit must control its own power in a decentralized control manner. The loads are connected to the MV feeders via Y/ $\Delta$  transformers. In this paper, it is assumed that the microgrid system operates in the islanded mode. Therefore, the DG units are responsible for compensating the reactive power of loads. The microgrid parameters are given in Table I.

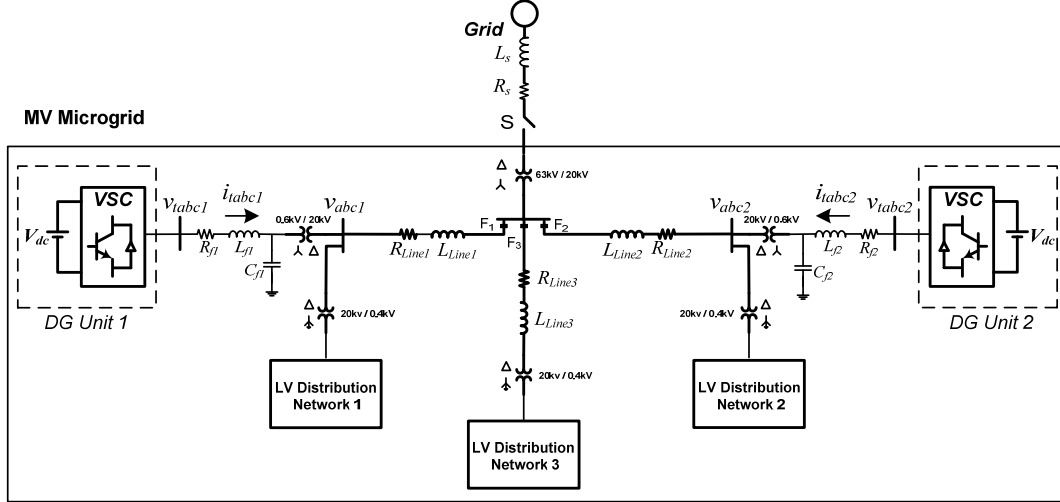


Figure 1. MV microgrid consisting of two DG units

TABLE I: Microgrid system parameters

| Parameter                      | Value                 | Comments                 |
|--------------------------------|-----------------------|--------------------------|
| $S_{base}$                     | 2.5 MVA               | DG ratings               |
| $Z_{line1}$                    | $0.7 + j 1.57 \Omega$ | 5.7 km overhead line     |
| $Z_{line2}$                    | $0.5 + j 1.25 \Omega$ | 4 km overhead line       |
| $Z_{line3}$                    | $0.1 + j 0 \Omega$    | 2 km underground cable   |
| $L_{f1}, L_{f2}$               | 0.3 mH                | series filter inductance |
| $R_{f1}, R_{f2}$               | 0.0015 $\Omega$       | series filter resistance |
| $C_{f1}, C_{f2}$               | 2200 $\mu F$          | filter capacitance       |
| $V_{dc}$                       | 1500 V                | dc bus voltage           |
| $f_s$                          | 2 kHz                 | switching frequency      |
| $P_{max_{DG1}}, P_{max_{DG2}}$ | 2 MW                  |                          |
| $Q_{max_{DG1}}, Q_{max_{DG2}}$ | 1.5 MVar              |                          |

Fig. 2 shows the block diagram of the control system for each DG unit. The control strategy comprises (i) a voltage control, and (ii) a power sharing control loop. The purpose of the voltage control loop is to keep the voltage of each bus of the microgrid within the acceptable limits.

As shown in [8], since the matrix transfer function of the DG subsystem in the  $\alpha\beta$ -frame is diagonal, two identical SISO controllers can be independently designed for the quadrature axes  $\alpha$  and  $\beta$ . In the  $\alpha\beta$ -frame, the reference signals are sinusoidal, and therefore, a PR controller is designed to ensure the excellent reference tracking [8], [9]. The reference signals are determined by an adaptive droop control strategy whose features will be explained in the next section.

The instantaneous real and reactive powers [10] are calculated and passed through a low pass filter (LPF) with the cutoff frequency of 5Hz. The LPF eliminates the ripples of power components. The averaged powers are then applied to the proposed droop controller to calculate the frequency and voltage magnitude of the reference signals in the  $\alpha\beta$ -frame. The reference generator unit produces the reference signals for the PR controller. The outputs of the voltage controllers are considered as the reference signals for the current controllers. These signals are compared with the currents of the series filter and are added to the

feedforward signals  $i_{o\alpha}$  and  $i_{o\beta}$ . The resultant signals are then applied to the current controllers to generate the control signals  $U_\alpha$  and  $U_\beta$ .

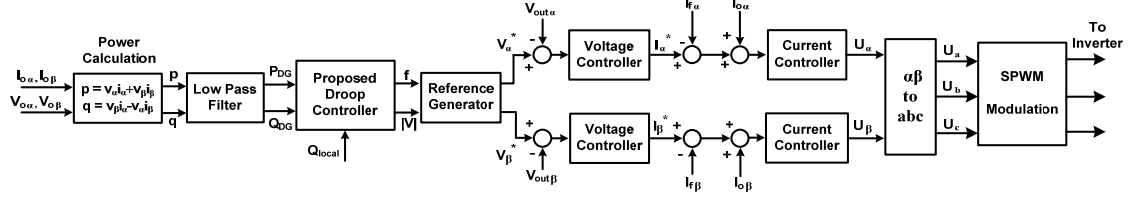


Figure 2. Structure of control system for each DG unit

### 3. PROPOSED METHOD FOR REACTIVE POWER COMPENSATION

Fig. 3 illustrates the proposed idea of the adaptive droop method for local reactive power compensation. Each DG unit measures its output reactive power and the reactive power consumed/generated by the downstream MV radial grid. For example, in Fig. 1, DG1 measures the reactive power of LV network 1 as a downstream MV grid. The downstream MV grid of a DG unit may contain several DG units in radial configuration. The droop characteristic curve is dynamically changed based on the reactive power of the local load, as given by:

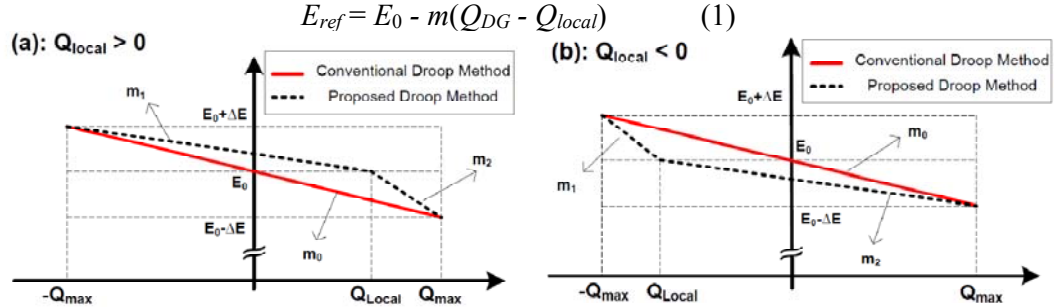


Figure 3. Proposed adaptive droop characteristics for local reactive power compensation, (a)  $Q_{local} > 0$  and (b)  $Q_{local} < 0$ .

In (1),  $E_0$ ,  $Q_{DG}$ , and  $Q_{local}$  denote the nominal voltage of DG unit, output reactive power, reactive power of the local loads located downstream of DG, respectively, and  $m$  is defined by:

$$m = \begin{cases} \frac{\Delta E}{Q_{max} - Q_{local}} & \text{if } Q_{DG} > Q_{local} \\ \frac{\Delta E}{Q_{max} + Q_{local}} & \text{if } Q_{DG} < Q_{local} \end{cases} \quad (2)$$

In (2),  $Q_{max}$  is the maximum reactive power that each DG unit can inject to the grid, and  $\Delta E$  is the voltage deviation whose limit is assumed 10% of the nominal voltage. When the local load is inductive, the center of the characteristic curve is moved to the right as shown in Fig. 2(a). In this case, the available capacity of the DG unit for reactive power compensation is reduced to  $Q_{max} - Q_{local}$  by increasing the slope  $m$  of the droop curve. Thus, the capability of capacitive reactive power compensation is increased to  $Q_{max} + Q_{local}$ . For capacitive local loads, the droop characteristic is changed according to Fig. 2(b) to increase the reactive power compensation capability. Stability analysis of the system can be performed based on the small

signal model and Eigenvalue analysis. The analysis shows that the system is robustly stable for all values of loads reactive power. To preserve the stability of droop method, the maximum value of parameter  $m$  in eq. (2) should be limited to  $\frac{10 \cdot \Delta E}{Q_{max}}$ .

#### 4. SIMULATION RESULTS

The islanded microgrid system of Fig. 1 is a 20-kV three-feeder distribution subsystem. Two electronically-coupled DG units are connected to Feeders 1 and 2 through step-up transformers. The DG units have the same parameters and they both use the proposed droop method for local reactive power compensation. The system parameters are summarized in Table I. To verify the performance of the proposed method, three simulation case studies are carried out using PSCAD/EMTDC software. To show the main advantages of the proposed droop method, the conventional droop controller is also simulated.

##### A. Case Study 1: Inductive Load Switching at Nonlocal Feeder

At  $t=2.5$  s, a sudden load change of +0.6 MVar is imposed to the LV side of feeder F3. Fig. 4 shows the reactive power of the three feeders. Fig. 5 shows the dynamic response of the microgrid system to this load change when the conventional droop controller and the proposed adaptive droop are used. When the conventional droop controller is used, the reactive power sharing between the DG units is performed without considering the local reactive load. Each DG unit compensates the reactive power of the loads according to its capacity. However, in the proposed scheme, before the load change, each DG unit compensates the reactive power of its dedicated load, and the reactive power flow in MV lines is almost zero as shown in Fig. 5(b, right column).

After the load change, the DG units share the demanded reactive power based on their dynamic droop characteristics. The adaptive droop characteristics of the DG units are not identical as each DG unit supplies different local load. Thus, as shown in Fig. 5(b, right column), the reactive power increase of the two DG units are not equal.

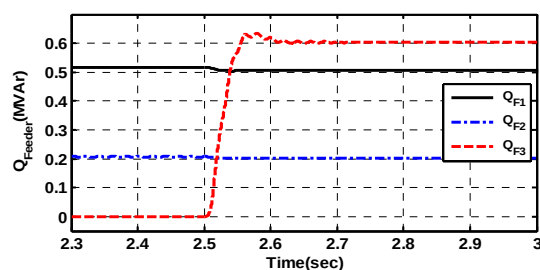


Figure 4. Reactive powers of feeders when a sudden load change of +0.6 MVar is imposed at LV side of feeder F3

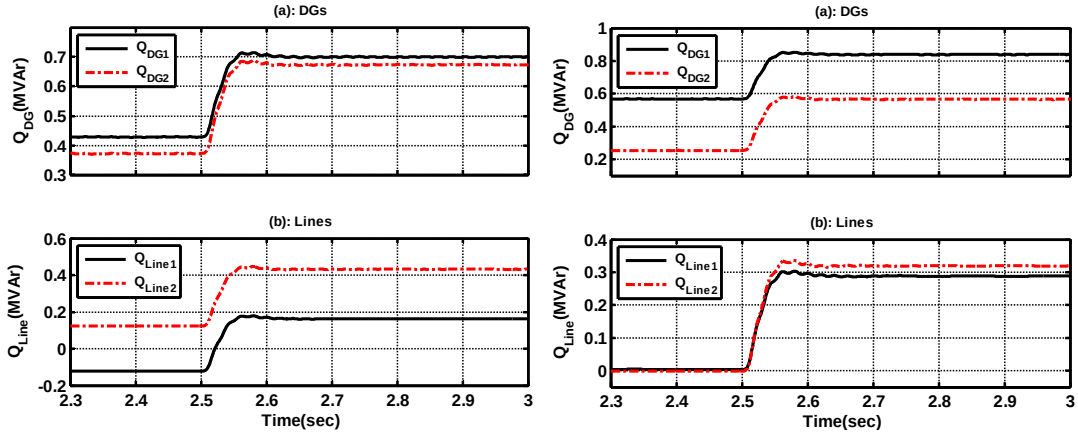


Figure 5. Dynamic response of the microgrid system to a load change in feeder F3: Reactive power components of, (a) DG units, and (b) lines (left: by using conventional droop controller, right: by using proposed adaptive droop controller)

## B. Case Study 2: Inductive Load Switching at Local Feeder

In this case study, a sudden inductive load change of +0.5 MVar is applied at the terminals of DG1 at  $t=2.5$  s. Fig. 6 shows the reactive power of the feeders due to this load change. Fig. 7, left column, shows the reactive powers of the DGs and lines when the conventional droop controller is used. Fig. 7(a, right column) shows that the reactive power of feeder F1 is completely compensated by DG1, whereas DG2 does not inject any reactive power due to this change. It should be noted that, after the load change, the reactive power sharing of feeder 2 is done based on the new adaptive droop characteristics. As observed from Fig. 7(b, right column), the reactive power sharing of DG2 for compensating feeder F2 increases since the inductive capacity of DG2 becomes more than that of DG1.

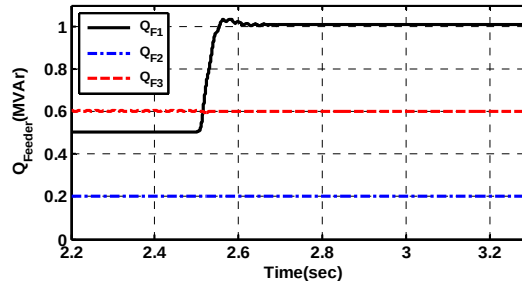


Figure 6. Reactive power components of loads when a sudden load change of +0.5 MVar is imposed at LV side of feeder F1

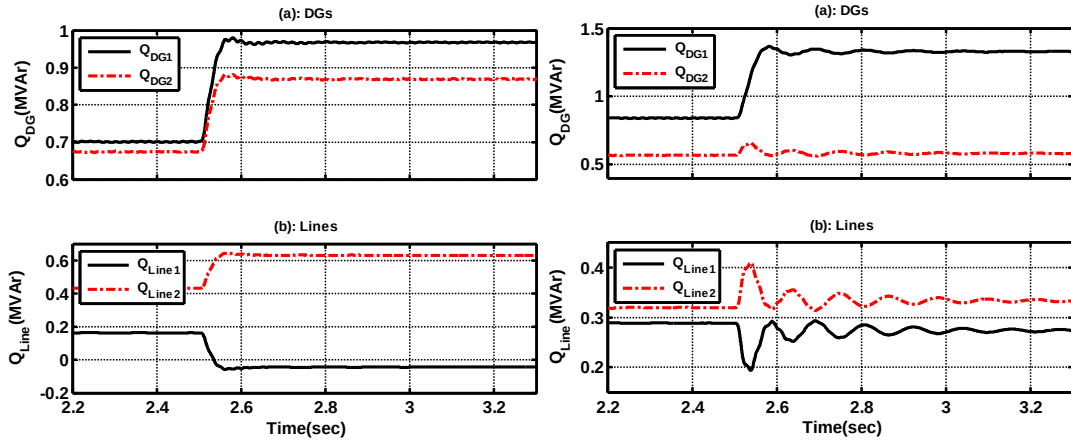


Figure 7. Dynamic response of microgrid system for Case 2: Reactive powers of, (a) DG units, and (b) lines (left column: conventional droop controller, right column: proposed adaptive droop controller).

### C. Case Study 3: Capacitive Load Switching at Nonlocal Feeder

In this study, a sudden capacitive load switching of -0.5 MVar is applied at the LV side of feeder F3 at  $t=2.5$  s. Fig. 8 shows the reactive powers of the feeders. Fig. 9(a, right column) shows that prior to this change, the reactive powers of feeders F1 and F2 are completely compensated by the DG units. Subsequent to the capacitive load change in feeder F3, the DG units share the demanded reactive power based on the dynamic droop characteristics of the DG units.

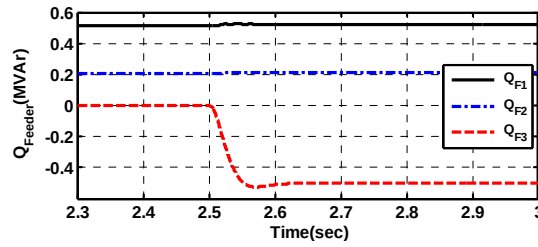


Figure 8. Reactive power components of loads when the capacitive load change of -0.5 MVar is imposed at LV side of feeder F3

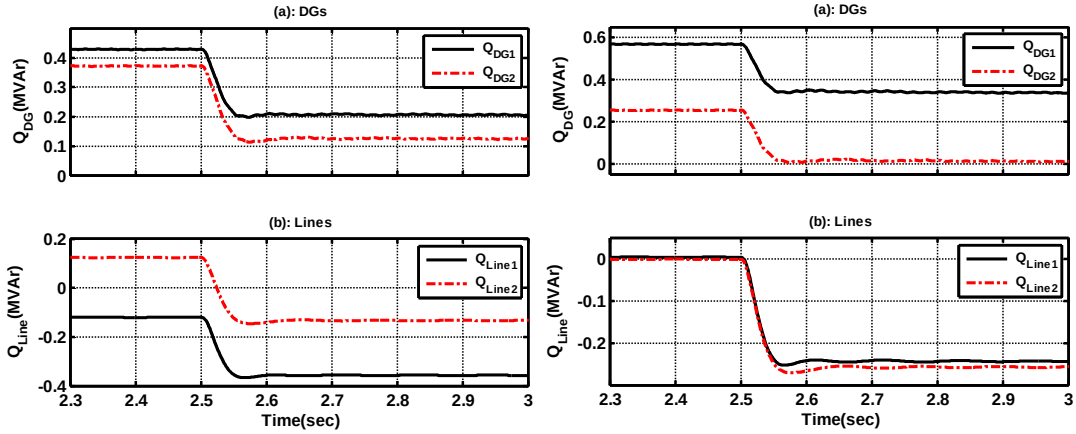


Figure 9. Dynamic response of the microgrid system to a load change in feeder F3: Reactive power components of, (a) DG units, and (b) lines (left column: conventional droop controller, right column: proposed adaptive droop controller).

## A. CONCLUSION

This paper presents a new control strategy for a multi-bus MG consisting of dispatchable electronically-coupled DG units. In the proposed method, the reactive power of a local load is completely compensated by its dedicated DG unit. However, the reactive power of nonlocal loads are shared among the adjacent DG units. The proposed adaptive droop scheme can be integrated with the existing voltage control methods. The droop characteristic curve is dynamically changed based on the reactive power of the local load. The performance of the proposed control strategy is verified by digital time-domain simulation studies in the PSCAD/EMTDC software. The simulation results show that the proposed strategy:

- is able to share the reactive power components among the DG units, and
- effectively compensates the reactive powers of the local loads such that the reactive power flow in the MV lines is minimized.

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